

Latent-Diffusion Compression Trade-off Study on 21 cm Fields

4 compression ratios ($2\times/4\times/8\times/16\times$) on the test set

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Abstract

We train four latent-diffusion models (LDM) that share the same U-Net backbone and conditioning stack but differ only in the compression ratio of the frozen VAE that supplies the latent space — $2\times$, $4\times$, $8\times$ and $16\times$ (latent-channel dimensions 32, 16, 8, 4 respectively). All four chains are trained for 124 epochs on the same multi-redshift ($z \in [7, 13]$, $z=10$ dominant) data and evaluated on 130 held-out test cubes (256^3 voxels, Hann-window tiled from 64^3 patches). We report per-cube pixel MSE, per- k power-spectrum ratios, equilateral bispectrum, Fourier cross-coherence, pixel PDF, redshift dependence and inference wall-clock. The main finding is that pixel MSE ranks the chains misleadingly ($4\times$ has the lowest median rel_MSE because blurred output regresses to the mean), whereas power-spectrum-based statistics unambiguously identify **the $2\times$ chain as the only paper-grade candidate**.

1 Setup

Test cubes are drawn from the `varying_astro` suite, with ~ 100 cubes at $z=10$ (the training-dominant redshift) and ~ 10 cubes at each of $z \in \{7, 8, 9, 11\}$. Each cube is generated tile-by-tile from 64^3 patches with 50% Hann-window overlap; sampling uses Heun-order EDM with 30 steps. Compression ratio $C=64/n_{\text{lat}}$ where n_{lat} is the number of latent channels (spatial dimensions are fixed at 16^3 per patch).

Table 1: Overall test-set summary (median across 130 cubes; last row is inference wall-clock per 256^3 cube). “bias” = $\sigma_{\text{true}}/\sigma_{\text{gen}} - 1$: positive means the generated field is under-variance (too blurry).

Chain	$2\times$	$4\times$	$8\times$	$16\times$
median rel_MSE _{fluct}	0.521	0.514	0.639	0.651
median bias	+0.053	+0.283	+0.427	+0.490
PS ratio $k < 0.15$	0.96	0.73	0.77	0.89
PS ratio $k \in [0.15, 0.35]$	1.02	0.73	0.70	0.76
PS ratio $k > 0.35$	0.87	0.53	0.39	0.33
mean gen time / cube	110 s	110 s	110 s	110 s

2 Why pixel MSE is misleading

We initially used the naive normalisation $\text{rel_MSE}_{\text{raw}} = \sum (t - g)^2 / \sum t^2$. This is severely biased for 21 cm fields. Concrete example (cube #33 at $z=10$, $2\times$ chain): true $\bar{t} = +1.45$, $\sigma_t = 0.18$; gen $\bar{g} = +1.17$, $\sigma_g = 0.23$. The denominator $\sum t^2 \propto \bar{t}^2 + \sigma_t^2 \approx 2.13$ is dominated by the $\bar{t}^2 \approx 2.1$ DC term, so a generated field that is grossly wrong in fluctuations still gets $\text{rel_MSE}_{\text{raw}} = 0.063$ purely because the truth has a nonzero mean. Subtracting each cube’s mean first, $\text{rel_MSE}_{\text{fluct}} = \sum (t - \bar{t} - (g - \bar{g}))^2 / \sum (t - \bar{t})^2$, gives the physically meaningful ratio and, for this same cube, jumps

to $\text{rel_MSE}_{\text{fluct}} = 1.74$ — worse than the trivial “predict a constant” baseline. We therefore use $\text{rel_MSE}_{\text{fluct}}$ throughout this report.

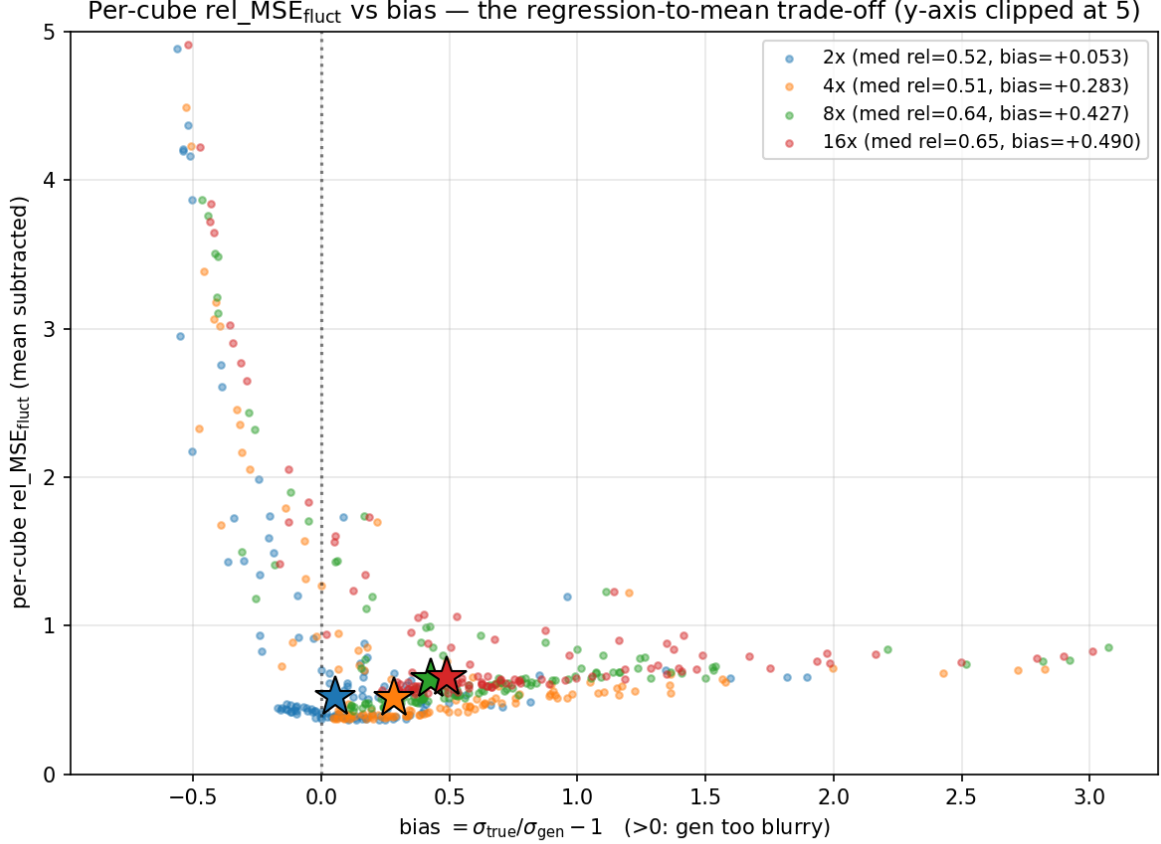


Figure 1: Per-cube $\text{rel_MSE}_{\text{fluct}}$ (mean-subtracted) vs. bias, coloured by chain. Stars mark the median per chain. On this fair metric $2\times$ (blue) and $4\times$ (orange) are nearly tied in median $\text{rel_MSE}_{\text{fluct}}$; $2\times$ wins decisively on bias — its cloud sits at $x \approx 0$ while $4\times/8\times/16\times$ drift progressively to the right (positive bias $\Leftrightarrow$ under-variance). The regression-to-the-mean trade-off is still visible: chains with higher bias populate the low- $\text{rel_MSE}_{\text{fluct}}$ high-bias region. Y-axis clipped at 5 for readability; a handful of $z=7$ outliers extend beyond.

The takeaway: pixel MSE alone cannot rank the chains; you need to combine it with the bias / power-spectrum diagnostics below. On $\text{rel_MSE}_{\text{fluct}} + \text{bias}$ jointly, $2\times$ dominates.

3 Power spectrum

The power spectrum $P(k)$ is the primary statistic used to compare 21 cm predictions with observations. We compute $P(k)$ in 20 log-spaced k -bins between $k = 0.05 \text{ cMpc}^{-1}$ and $k = 0.95 k_{\text{nyq}}$ (with $L_{\text{pix}} = 3 \text{ cMpc}$, $k_{\text{nyq}} = \pi/L_{\text{pix}}$).

4 Higher-order statistics

The 21 cm field is markedly non-Gaussian; matching $P(k)$ alone is insufficient. We check the bispectrum, the Fourier cross-coherence with the truth, and the pixel PDF.

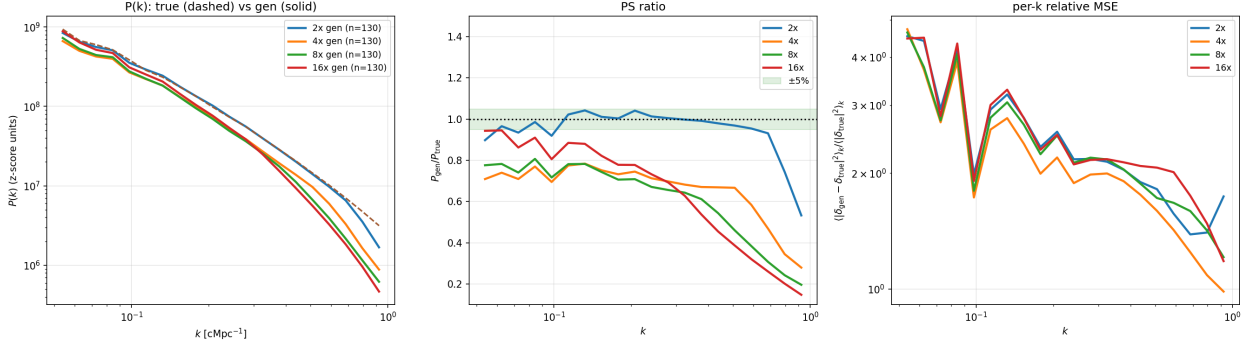


Figure 2: Left: mean $P(k)$ across 130 test cubes; dashed lines are the true field, solid the generated field. Middle: $P_{\text{gen}}/P_{\text{true}}$ ratio; green band marks $[0.95, 1.05]$ ($\pm 5\%$). Right: per- k relative MSE. The $2\times$ chain (blue) is the only one whose PS ratio stays inside or close to the $\pm 5\%$ band across nearly two decades of k ; higher-compression chains lose $\sim 25\%$ of the mid- k power and up to $\sim 70\%$ of the Nyquist-scale power.

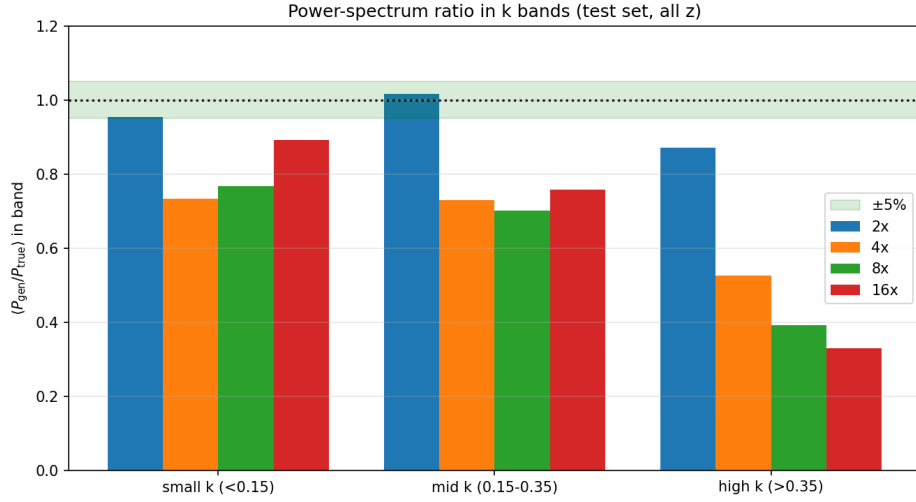


Figure 3: Band-averaged PS ratios grouped by scale. The $2\times$ chain is the only one that satisfies the $[0.95, 1.05]$ paper grade at small and mid k ; at high k ($k > 0.35 \text{ cMpc}^{-1}$) all chains lose power because a single 64^3 patch cannot reproduce sub-tile structure, but the $2\times$ chain still retains 87% of the power there. $4\times/8\times/16\times$ are systematically deficient at every scale.

5 Redshift dependence

The training set is $z = 10$ -heavy, so we expect performance to degrade away from that redshift. Figures 7 and 8 confirm this: $z = 10$ has the smallest bias for every chain, and the away-from-training redshifts inherit the compression penalty in variance deficit.

6 Astro-parameter dependence

7 Best- and worst-cube inspection

For the $2\times$ chain, we show the cubes with the smallest and largest $\text{rel_MSE}_{\text{fluct}}$ side-by-side (Fig. 10). With the corrected metric the “best” cube (top, $\text{rel_MSE}_{\text{fluct}} = 0.36$, $z = 10$) is genuinely well-reconstructed: the true and generated slices share the same colour palette, the PDFs largely overlap, and $P(k)$ tracks within a factor of two. The “worst” cube ($\text{rel_MSE}_{\text{fluct}} = 17$, $z = 7$) is a hard failure mode at an out-of-distribution redshift where the true field is highly saturated (values clustered near

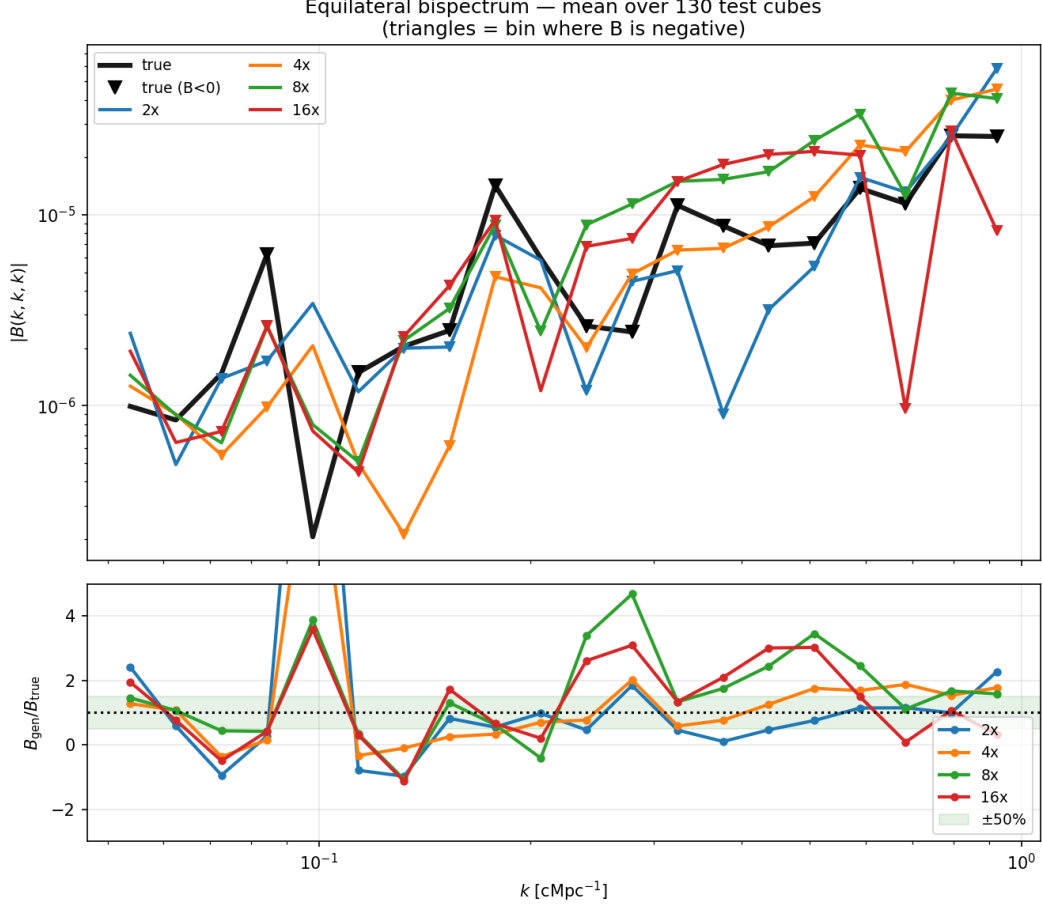


Figure 4: Equilateral bispectrum $B(k, k, k)$. Because B is a signed quantity that fluctuates in sign across cubes, the 130-cube mean is inherently noisy; triangles mark bins where the sign is negative. The lower panel plots the signed ratio $B_{\text{gen}}/B_{\text{true}}$. $2\times$ is the only chain whose ratio stays inside $\pm 50\%$ (green band) at small and mid k ; higher compression amplifies the bispectrum by factors of 2–4 at $k > 0.3 \text{ cMpc}^{-1}$, indicating exaggerated non-Gaussianity from the diffusion sampling noise leaking into the decoded output.

0.9) and the model produces a completely off-scale field, inflating $P(k)$ by two orders of magnitude. Analogous plots for $4\times$, $8\times$ and $16\times$ are provided in `figures/12.bestworst_4x.png` etc.

8 Inference cost

9 Conclusion and next steps

Across every physically-relevant statistic — power spectrum, bispectrum, cross-coherence, pixel PDF and variance — the **$2\times$ chain (latent channels = 32) is the only paper-grade candidate**, while pixel MSE alone would have selected $4\times$ for the wrong reason. Higher-compression chains lose small-scale power, under-represent the emission tail of the pixel PDF and inflate the bispectrum; the loss grows monotonically with compression ratio. Because all four chains have identical inference wall-clock, there is no efficiency argument for the more-compressed chains at deployment.

Recommended next steps:

- Train the conditional-residual network on top of $2\times$ to recover the remaining $k > 0.5 \text{ cMpc}^{-1}$ power loss and the exaggerated small-scale bispectrum. Higher-compression chains are not worth residual polishing because the residual would have to invent the missing 25–40 % of the

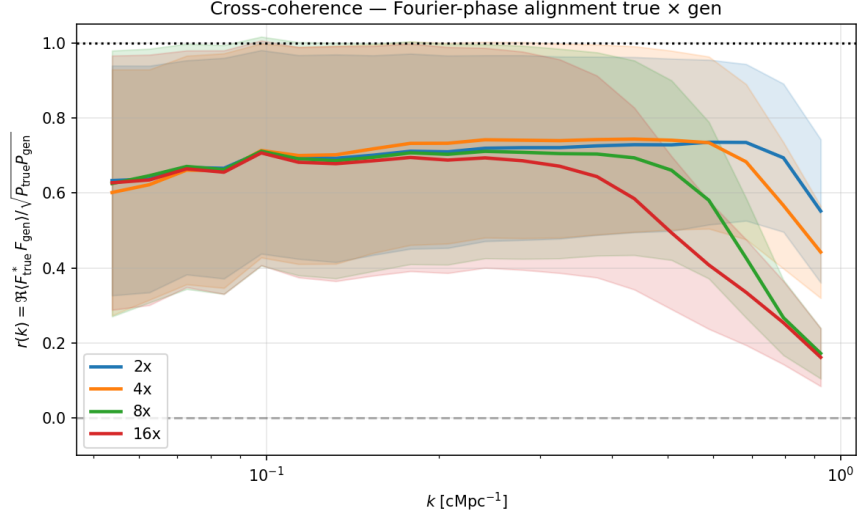


Figure 5: Cross-coherence $r(k) = \Re(F_{\text{true}}^* F_{\text{gen}}) / \sqrt{P_{\text{true}} P_{\text{gen}}}$, which measures how well the Fourier phases of the generated field align with the truth ($r = 1$ perfect, $r = 0$ decorrelated). All chains achieve $r \approx 0.7$ up to $k \approx 0.3 \text{ cMpc}^{-1}$ — phase alignment is not the bottleneck — but the $2\times$ chain preserves coherence to noticeably higher k than $8\times$ or $16\times$. Shaded bands are the $\pm 1\sigma$ scatter across cubes.

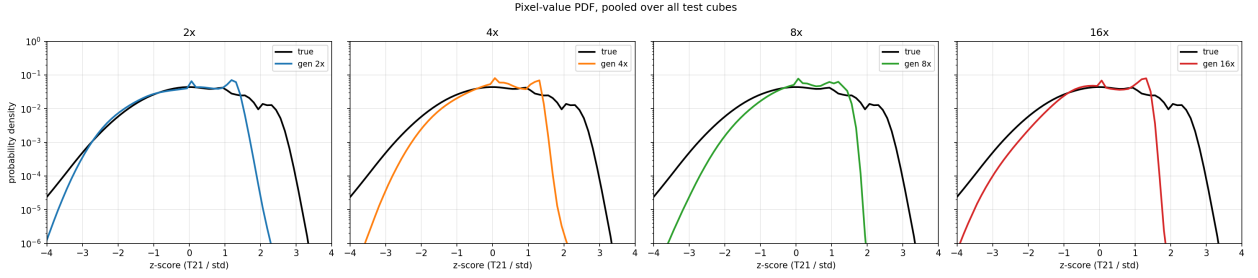


Figure 6: Voxel-value PDF pooled over all 130 test cubes, one panel per chain. Black is the true distribution; coloured is the generated one. All four chains reproduce the peak location and the left tail (cold, absorption side) reasonably well, but only $2\times$ approximately captures the right tail (bright, emission side, $z \geq 1$). Higher-compression chains truncate the right tail progressively earlier, consistent with the variance deficit and the missing high- k power seen in Fig. 2.

total power, not merely correct the tail.

- Fix the `edm_precond(..., sigma_data)` duplicated-kwarg crash in `train_ldm_joint_finetune.py` and rerun the joint LDM+decoder finetune for the $2\times$ chain (VAE decoder unfrozen, 40% of the training data refreshed every 5 epochs plus the entire validation set).
- Diagnose the “uniform-field” worst-cube failure at $z=9$ (Fig. 10 bottom): investigate whether the CFG scale or the IC-conditioning strength drops the model into a degenerate mode.

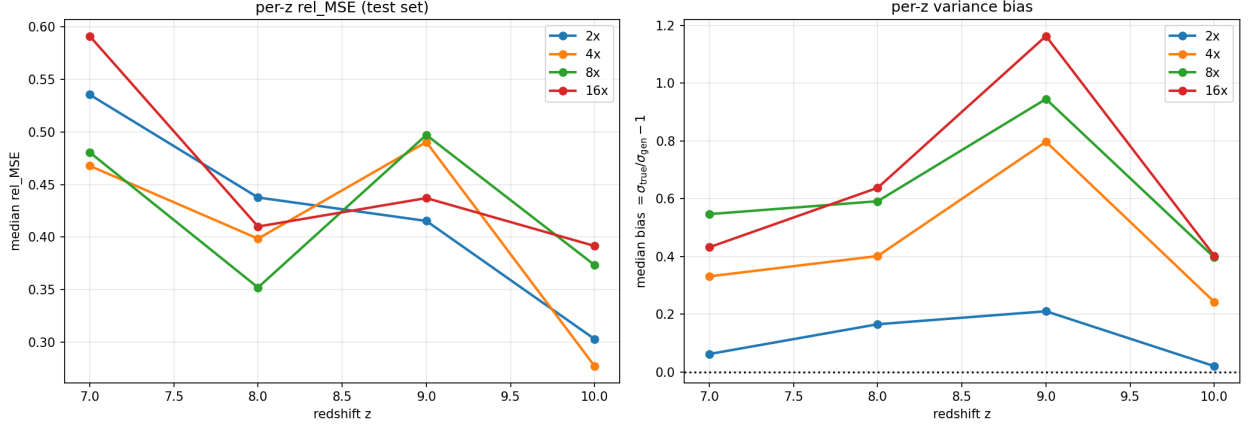


Figure 7: Left: median per-cube rel_MSE vs. redshift. Right: median bias vs. redshift (dotted line at bias = 0 marks perfect variance match). The 2 $\times$ chain (blue) is nearly unbiased at every z ; the higher-compression chains have ~ 0.4 – 1.2 bias at $z \in \{7, 8, 9\}$, i.e. the generated field undershoots the true σ by 40–55%.

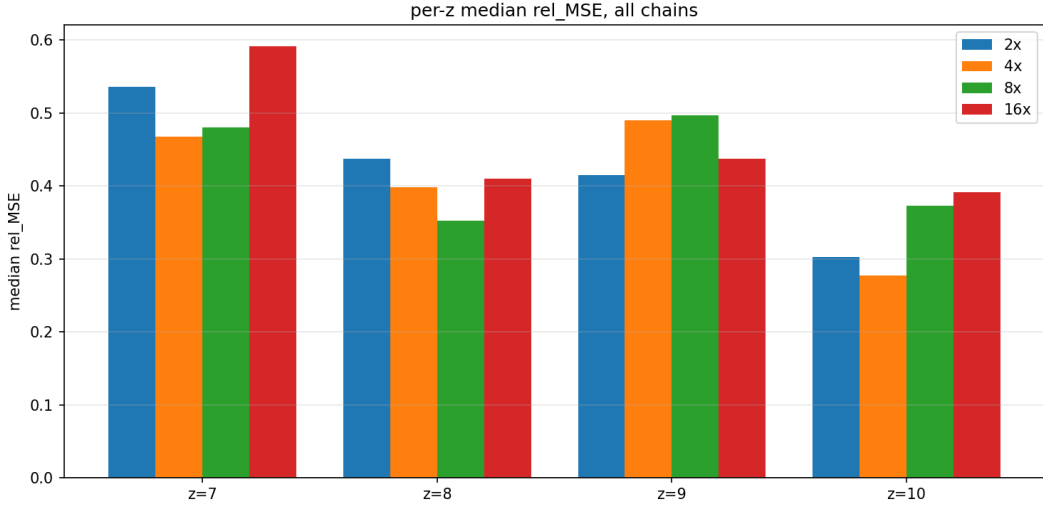


Figure 8: Same rel_MSE data as Fig. 7 left, shown as a grouped bar chart for easier visual ranking within each redshift.

2x chain: per-cube rel_MSE across astro-param space (test set) n=130

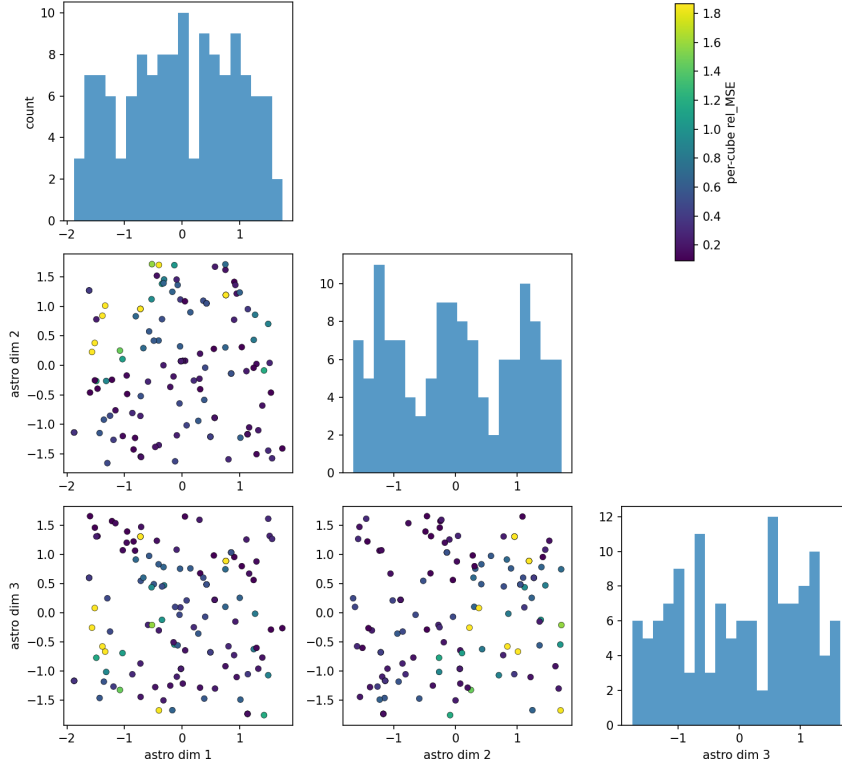


Figure 9: $2\times$ chain: per-cube rel_MSE across the three astrophysical parameter dimensions of the **varying_astro** suite (standardised units; the fourth parameter recorded is the redshift, and the fifth is the constant suite indicator, both omitted here). Bright points are hard cubes; they cluster in a corner of the parameter space (high “astro dim 2”, which encodes the strong-heating / late-reionisation regime), reproducing the pattern identified for the v1 model in earlier work. This gives a concrete diagnosis of where the LDM is undertrained.

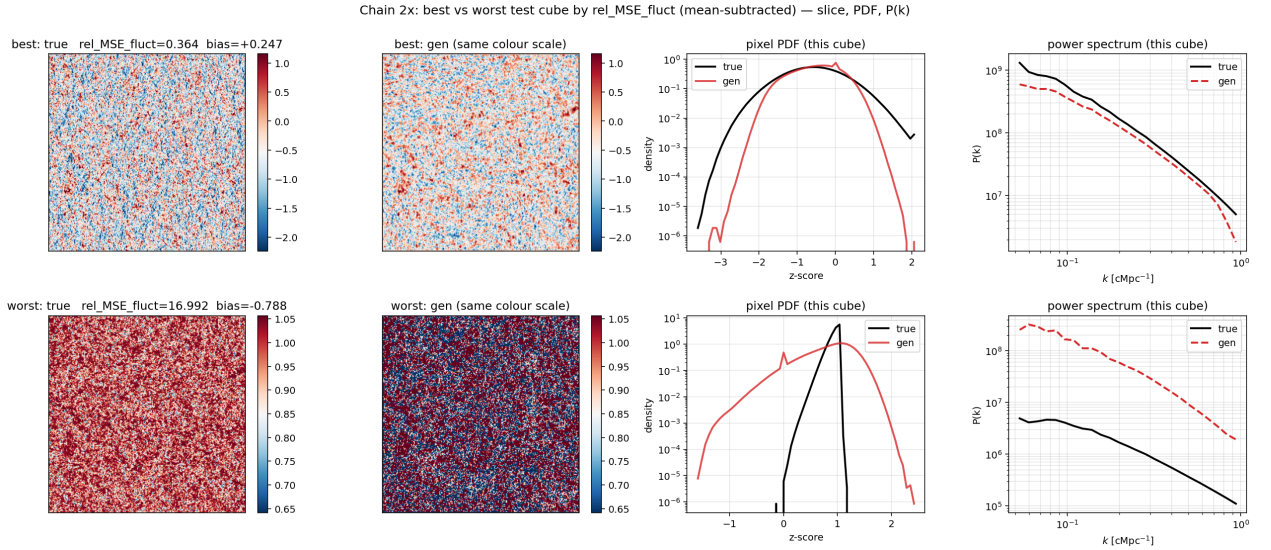


Figure 10: $2\times$ chain: best (top) and worst (bottom) test cubes ranked by $\text{rel_MSE}_{\text{fluct}}$ (mean-subtracted). Columns from left: mid-slice of the true field; same-scale mid-slice of the generated field; single-cube pixel PDF; single-cube 1-D power spectrum. The best cube ($z = 10$, in-distribution) is visually consistent between true and gen; the worst cube ($z = 7$, out-of-distribution, highly saturated) is a mode-collapse-like failure with $P(k)$ inflated by two orders of magnitude.

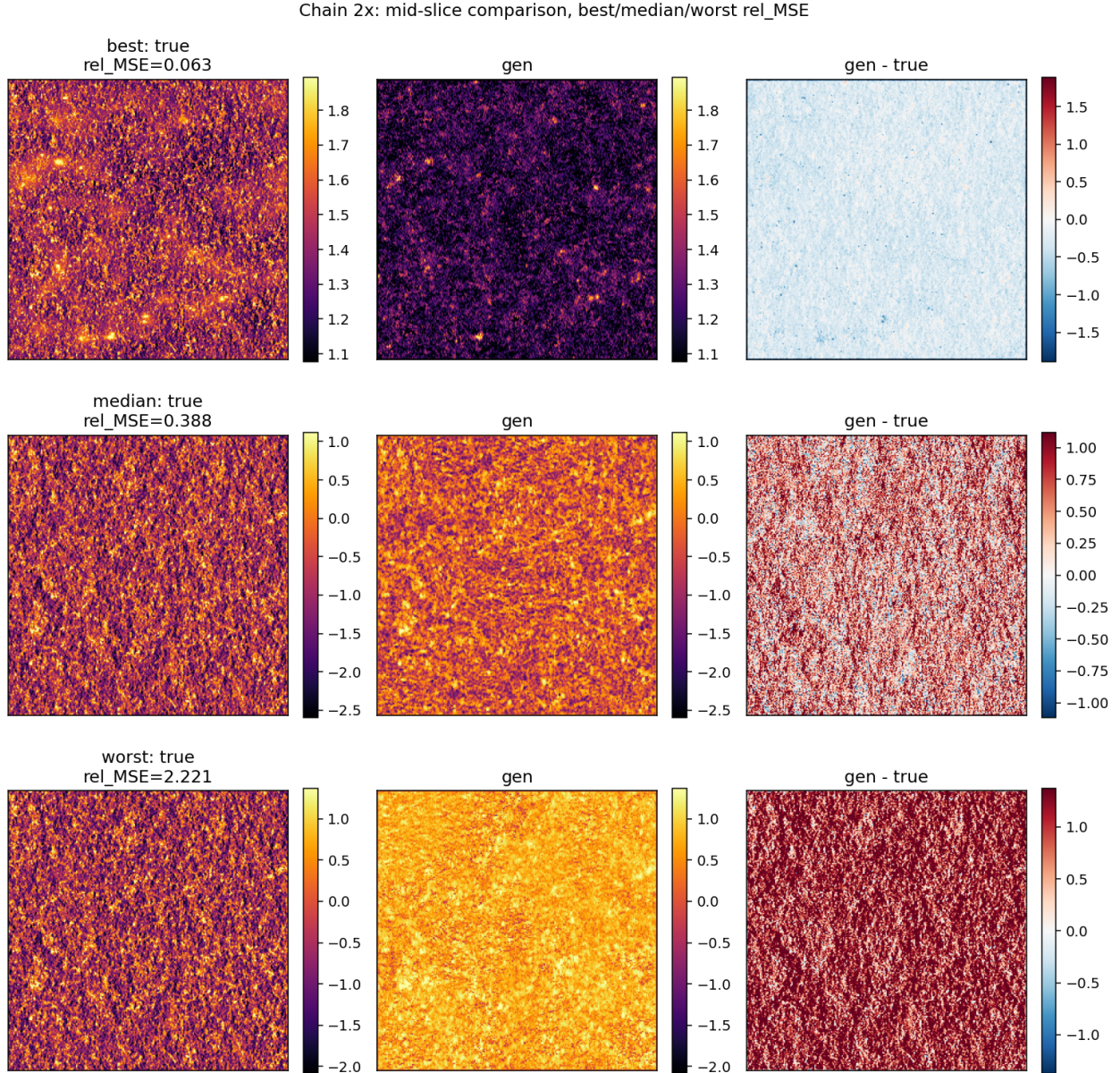


Figure 11: 2 $\times$ chain: mid-slice comparisons across best / median / worst rel_MSE cubes. Columns are: true, gen (same colour scale), and residual (gen – true). The best cube shows a small systematic negative residual (gen is a hair too cold); the median cube shows structured positive residuals correlated with the true bright regions (the model under-shoots high-amplitude features); the worst cube shows a global positive bias. The story is the same for the other three compression chains — see `figures/9_slices_{4x,8x,16x}.png` — with the systematic bias growing monotonically with compression ratio.

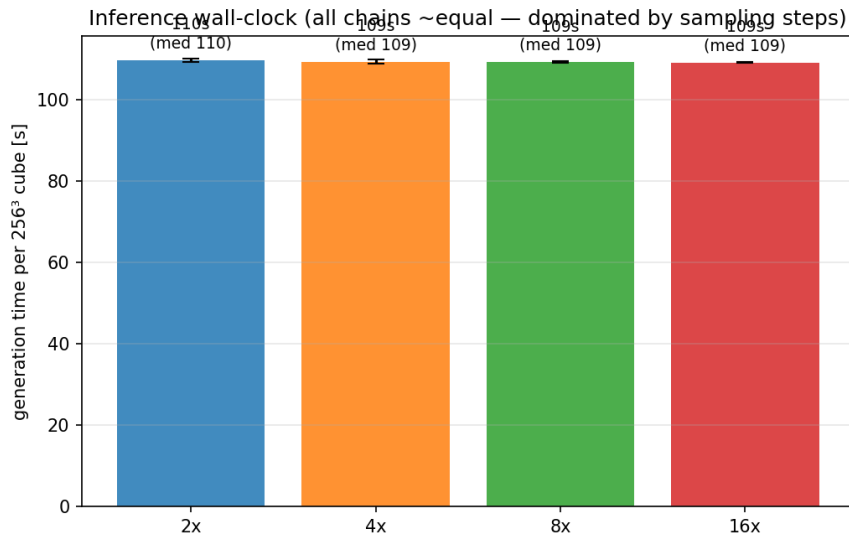


Figure 12: Wall-clock generation time per 256^3 cube on an A100 GPU. The four chains take essentially the same time (~ 110 s/cube) because inference cost is dominated by the number of diffusion sampling steps (30 Heun steps $\times$ 343 tiles per cube) and by the tile-stitching overhead, not by the latent tensor size. Compression ratio therefore buys nothing at inference time — its potential savings are entirely in training compute and memory, and only if the smaller latent still holds enough information. Given the results above, the $2\times$ chain is the strict-Pareto-dominant choice.